

Unified Dynamic Systems Optimization Engine (UDSOE)

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Abstract

This document outlines the vision for the **Unified Dynamic Systems Optimization Engine (UDSOE)**, a from-scratch computational framework designed to model and simulate multi-agent systems under constraints. The framework integrates numerical solvers, optimization methods, and agent-based utility modeling into a unified architecture, emphasizing transparency, mathematical rigor, and modularity for research purposes.

1 Motivation

Real-world systems often combine physical dynamics, multi-agent interactions, and constraints. Current tools are fragmented: physics engines simulate motion, economic models simulate agent behavior, and optimization solvers focus on mathematical objectives. No single framework unifies these aspects in a transparent and conceptually rigorous way. **UDSOE addresses this gap** by providing a modular environment to study and optimize complex systems, enabling experimentation across physics, economics, and constrained optimization.

2 Objective

The primary objectives of UDSOE are:

- Simulate the evolution of multi-agent systems under constraints
- Integrate agent utility modeling with system dynamics
- Implement numerical solvers and optimization methods from scratch for full transparency
- Provide a modular, research-oriented framework rather than a commercial tool

3 Architectural Philosophy

A from-scratch implementation ensures complete understanding of numerical methods, stability, and error propagation. The project emphasizes conceptual clarity and mathematical rigor over computational performance. By manually building core modules, the framework demonstrates mastery of linear algebra, differential equations, and constrained optimization, while maintaining modularity for future extensions. Transparency in the numerical pipeline allows clear analysis of system behavior and optimization results.

4 Capabilities and Limitations

Capabilities

- Differential equation solvers (Euler, Runge-Kutta 4)
- Gradient-based constrained optimization
- Multi-agent utility integration
- Transparent state-space modeling

Limitations

- Prioritizes conceptual clarity over industrial-scale computational speed
- Assumes differentiable dynamics for optimization
- Initial system scale is limited for tractability